Parth Deshmukh | Data Analyst

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SUMMARY

Data Analyst with 4 years of experience designing, implementing, and optimizing data solutions to drive actionable insights and operational efficiency. Skilled in SQL, Python, R, ETL pipelines, and cloud-based data platforms, with expertise in data visualization using Tableau and Power BI. Experienced in predictive modeling, machine learning, statistical analysis, and exploratory data analysis (EDA) to support data-driven decision-making. Strong track record in building scalable data pipelines, automating reporting, and translating complex datasets into clear, business-focused insights.

SKILLS

Programming & Databases: SQL (Oracle, SQL Server, Hive), Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R, VBA
Data Engineering: ETL workflows, Data pipeline development, Data integration, Query optimization, Database modeling, AWS (data storage & processing), Hive
Data Science & Analytics: Predictive modeling (regression, classification, clustering, time series), Machine learning, Statistical analysis, Hypothesis testing, A/B testing, Fraud detection models, Forecasting
Visualization & Reporting: Tableau, Power BI, Dashboard development, Data storytelling, automated reporting
Domain Knowledge: Healthcare claims and clinical data, Population health analytics, Risk factor analysis, Financial data analytics (QuickBooks, Credit Karma), Credit monitoring, Fraud prevention, Regulatory compliance (HIPAA, PHI)
Core Competencies: Exploratory Data Analysis (EDA), Business intelligence, Process automation, Data governance, Stakeholder communication, Data quality management

EXPERIENCE

Data Analyst | UnitedHealth Group | MA, USA **Jun 2024 – Present**

- Queried, cleaned, and merged 10M+ healthcare claims and clinical records using SQL, Python, and ETL pipelines, improving data accuracy and processing speed by 35%.
- Designed and automated Tableau and Power BI dashboards to track utilization trends, medical costs, and operational KPIs, cutting manual reporting time by 40%.
- Built and enhanced data pipelines and SQL queries across Oracle, SQL Server, and Hive, reducing query runtimes and enabling faster business reporting.
- Put into practice predictive modeling and machine learning techniques (regression, clustering, and classification) to forecast patient outcomes and identify fraud or cost-saving opportunities.
- Conducted exploratory data analysis (EDA) with Python and R to uncover patterns in claims and pharmacy data, driving insights for clinical and operational initiatives.
- Partnered with clinicians, actuaries, and executives to translate analytics into business strategies, directly supporting healthcare cost management and quality improvement programs.
- Ensured compliance with HIPAA & PHI regulations while standardizing governance practices for secure, enterprise-wide data use.

Data Analyst | Mphasis (Healthcare) | India **May 2021 – Jul 2023**

- Extracted, cleaned, and blended multi-source healthcare claims and patient datasets (5M+ records) using SQL, Python, and ETL workflows, improving data quality and accessibility by 30%.
- Designed and mechanized Power BI and Tableau dashboards to monitor utilization, claims processing, and patient risk factors, reducing manual reporting efforts by 40%.
- Built and streamlined data pipelines and database structures across SQL Server and AWS, improving processing efficiency and query performance by 25%.
- Utilized predictive analytics techniques (regression, clustering, and forecasting) in Python/R to identify high-risk patients, improve wellness program targeting, and support cost-control initiatives.
- Streamlined claims reconciliation and billing workflows through automation, enhancing accuracy and reducing processing errors by 20%.
- Partnered with cross-functional teams to translate analytical insights into operational strategies, directly improving patient engagement and resource allocation decisions.
- Ensured compliance with HIPAA-aligned data governance and security standards, safeguarding sensitive patient and insurance information.

Data Analyst | Intuit Financial | India **Mar 2020 – Apr 2021**

- Worked with large-scale financial datasets, using SQL and Python to clean, transform, and prepare data that refined accuracy and reduced errors by 25%.
- Built and computerized interactive dashboards in Tableau that gave team's real-time visibility into revenue trends, customer transactions, and credit utilization—cutting manual reporting time by 35%.
- Partnered with product and data science teams to create AI-driven financial insights in QuickBooks, helping small businesses forecast sales and profits more easily.
- Contributed to personalized financial recommendations for Credit Karma users, which enhanced engagement and boosted conversion rates for financial partners.
- Helped design and optimize data pipelines and ingestion processes, making it faster and easier for teams to access reliable financial data.
- Supported fraud detection and credit monitoring by applying statistical models, strengthening Intuit's ability to protect users and build customer trust.
- Shared findings and recommendations with business leaders, translating complex data into clear insights that shaped product improvements and customer experience strategies.

EDUCATION

Master of Science in Data Analytics Engineering | Northeastern University Boston, USA **Aug 2025**
Bachelor of Engineering | Savitribai Phule Pune University Pune, India **May 2023**

CERTIFICATIONS

- AWS Academy Cloud Foundations (Credentials)
- Engineering Design for a Circular Economy (Credentials)

PROJECTS

Diabetic Prediction model (GitHub)

- Expanded a machine learning pipeline for early diabetes detection; improved XGBoost model achieved 90% accuracy and 85% recall.
- Created interpretable logistic regression baseline to benchmark key risk factors and validate model structure.
- Implemented an SVM with Random Fourier Features for efficient RBF approximation and integrated regularized XGBoost with early stopping; delivered best recall (85%) and precision (88%).

Customer Churn Prediction (GitHub)

- Employed Machine Learning models on dataset by deploying Linear Regression, Logistic Regression, Decision Tree, Support Vector Machine achieving accuracy of 92.87%, and a comprehensive understanding of customer behavior
- Boosted marketing strategies based on customer segmentation, including loyalty programs, win-back campaigns, and directed promotions, reducing the churn rate.

EEG Classification Model (GitHub)

- Employed a Recurrent Neural Network on patient-specific seizure detection model using machine learning, focusing on EEG Time Series data in python.
- Engineered dropout regularization during training, leading to a 12% reduction in overfitting, and early stopping strategies, resulting in a 25% decrease in training time without sacrificing model accuracy.

Prediction of Loan Defaulter (GitHub)

- Executed extensive exploratory and statistical analysis to identify key variables, patterns, outliers, and missing values in the datasets.
- Utilized SMOTE techniques and KNN imputation recover 7.7 % of data to balance the dataset and missing values to improve machine learning model performance.
- Built and compared multiple classification models using Logistic regression, KNN, Random Forest, and Decision Trees.
- Compared multiple classification reports of all models and Random Forest performed with highest accuracy and recall of 90%

Temporal & Geospatial Crime Forecasting (GitHub)

- Detected 243 time-series outliers using Seasonal-Hybrid ESD; identified demographic bias via Chi-square tests to improve ETL accuracy.
- Performed geospatial hotspot clustering using GeoPandas and HDBSCAN; spatial autocorrelation (Moran's I = 0.23) enabled 15% improvement in patrol planning.
- Built a rolling-origin cross-validated ARIMA model (MAPE: 8.7%, RMSE: 15.4%); systematized 360-day forecast reports with dynamic seasonality adjustments and incorporated scheduling via Airflow to support monthly business planning.